

JINGTIAN WU

 (+1) 607-280-8173  jw2349@cornell.edu
 [jingtianwu.github.io](https://github.com/jingtianwu)  [Google Scholar](#)  [Github](#)  [LinkedIn](#)

EDUCATION

Cornell University

May 2026 (expected)

B.A. in Computer Science

GPA: 3.97/4.00

Related courses: NLP, RL, ML, HCI, CV, Algorithms, AI Foundations, AI Philosophy, Functional Programming, Linear Algebra, Discrete Math, Calculus, Computer Systems.

RESEARCH INTERESTS

My research interests include natural language processing, machine learning, and reinforcement learning. I am especially drawn to enhancing the **reasoning** and **real-world understanding** of large language models; designing **self-evolving models** capable of continual improvement; extending LLMs to **multimodal learning**; and exploring **interdisciplinary applications** across diverse domains.

RESEARCH EXPERIENCE

Behavioral Evaluation and Control in AI-Simulated Patients

Summer 2025 – Present

Advisor: Claire Cardie, Collaborator: Yann Hicke

Cornell University

- Investigating behavioral consistency in large language models used as AI-simulated patients for medical education.
- Developing methods to detect and control these behaviors using supervised classifiers, multi-agent systems, and reinforcement learning, enabling more realistic and pedagogically aligned simulations.

OmniWizz — Multimodal Inspirational Tool

Summer 2025 – Present

Advisors: Qi Yang and Gilly Leshed

Cornell University

- Addressed the need for a tool to inspire musical creativity by enabling seamless transformation across text, image, and audio, fostering cross-modal idea generation and exploration.
- Designed and implemented the full-stack architecture for a unified multimodal framework, delivering flexible input-output combinations to empower diverse and innovative creative workflows. [View Demo]

Automatic Synthetic Data Generation

Spring 2025

Advisor: Claire Cardie

Cornell University

- Exploring automated dataset construction as a step toward **self-evolving models**, focusing on **scalable synthetic data generation** for continual domain adaptation in medicine.
- Built a hierarchical knowledge map expansion pipeline that recursively generates 226 subfields and produces 2.9K+ expert-student dialogues and MCQA items without human annotation.
- Fine-tuned a compact student model, achieving **+6.76% accuracy over baseline** while using **6.4× fewer training samples** than supervised methods.

Culturally and Temporally Contextualized Lyric Generation

Spring 2025

Advisor: Matthew Wilkens

Cornell University

- Investigating how to adapt large language models for lyric generation that is culturally, temporally, and emotionally grounded.
- Curated a 31,000-song corpus and fine-tuned LLaMA 3.1 8B with LoRA for context-aware generation.
- Achieved **+67.4% ROUGE-2** and **+18.9% human ratings** over baseline systems.

Automatic Code Generation System

Advisors: Claire Cardie and Wenting Zhao

Fall 2024

Cornell University

- Tackled the challenge of enabling LLM-based code generation to autonomously complete complex, multi-step software engineering tasks.
- Built a multi-agent framework with automated task discovery, curriculum learning, progress monitoring, and rollback recovery, and evaluated it on the Commit-0 benchmark.

Enhancing Multi-Hop Reasoning in LLMs

Advisors: Claire Cardie and Wenting Zhao

Summer 2024

Cornell University

- Tackled LLM hallucinations and reasoning errors in multi-hop QA.
- Developed Reasoning Court, extending ReAct with two retrieval-reasoning agents and an LLM judge capable of selecting or synthesizing factually grounded, coherent answers.
- Outperformed few-shot prompting baselines on HotpotQA, FEVER, and MuSiQue, achieving up to +8.2% EM improvement while reducing LLM calls compared to hybrid baselines.

Computer Music Synthesis and Composition

Advisor: Roger B. Dannenberg

Summer 2023

Carnegie Mellon University (remote)

- Explored mapping the digits of π to the minor pentatonic scale to algorithmically compose music, simulating traditional Chinese instruments and producing an original countermelody.

Algorithms for Big Data

Advisor: David Woodruff

Summer 2023

Carnegie Mellon University (remote)

- Addressed the high computational cost of GAN-based cartoonization for resource-constrained devices by developing LWAnimeGAN, reduced channel sizes, and removal of HourGlass blocks, reducing parameters by 8.8%, FLOPs by 56.1%, and inference time by 18.6% while preserving visual quality.

RESEARCH-RELATED PROJECTS

LLM Framework for Academic Peer Review

Collaborator: Thorsten Joachims and Zhaolin Gao

Spring 2024

Cornell University

- Led full-stack development of the demonstration system for the research paper *Reviewer2: Optimizing Review Generation Through Prompt Generation*, optimizing the system for interactive presentation.
- Presented the system at BOOM 2024 (Bits On Our Minds), featured in the *Cornell Chronicle*.

PUBLICATIONS

- [1] **Jingtian Wu** and C. Cardie, *Reasoning court: Combining reasoning, action, and judgment for multi-hop reasoning*, *arXiv preprint arXiv:2504.09781*, 2025.
- [2] **Jingtian Wu**, “Algorithmic composition of music utilizing the digits of pi,” in *Highlights in Science, Engineering and Technology*, vol. 85, 2024, pp. 570–577.
- [3] X. Zhao, Y. Zhou, **Jingtian Wu**, Q. Xu, and Y. Zhang, “Turn real people into anime cartoonization,” in *Proc. IEEE ICCECE*, 2021, pp. 387–393. DOI: 10.1109/ICCECE51280.2021.9342433.

TEACHING EXPERIENCE

CS 3700: Foundations of AI Reasoning and Decision-Making, Cornell

Fall 2024, Fall 2025

SKILLS

Languages	Python, JavaScript, Java, OCaml, $\text{\LaTeX}$, Markdown, HTML/CSS
Web Development	Node.js, Flask, FastAPI, React
Frameworks/Libraries	PyTorch
Tools	Visual Studio Code, Jupyter, GitHub